

HANDOUTS

115.109

Introduction to Business Analytics

Topic 7:

Normal Distribution; Segmentation Difference in Means

Dr Andrew Murphy

(Please contact me via [Communication Tools in Stream](#))

Week (starting)	Business Problem	Data & Analysis	Chapter (Black et al)	Assessments
1 (13/7)	Understanding Data: discounts & margin	Calculating data in Excel; GST, discounts and Margins	1	Lab1
2 (20/7)	Operational Monitoring: Dashboards	Tabulating & Graphing	2	Lab2
3 (27/7)	Industry trends	Statistics NZ: retail survey Descriptive stats: averages	3	Lab3
4 (3/8)	Customer trends	Statistics NZ: census Descriptive stats: variation	3	Lab4
5 (10/8)	Transaction data: product sales	Transaction data sources. Sampling	7	Lab5
6 (17/8)	Loyalty cards	Loyalty card schemes. Probability; proportions, χ^2	4, 12	Lab6 Test 1
Mid	Semester	Study	Period	
7 (7/9)	Segmentation	Statistics NZ: census. Normal distribution; Means	6, 8	Lab7
8 (14/9)	Determining difference (1)	Means; ANOVA	9, 10, 11	Lab8
9 (21/9)	Associating and Predicting	Explaining relationships: Correlation & regression	13	Lab9
10 (28/9)	Determining Difference (2)	Explaining relationships (continued)	14	-
11 (5/10)	Trends over time	Time series, Indices Unstructured data: social media	15	Lab11 Test 2
12 (12/10)	Corporate reporting	Finance Dashboards reporting		Report

Assessments Schedule

2026	Assessment	Lab 1&2	Lab 3&4	Lab 5&6	Test 1	Lab 7&8	Lab 9,11	Test 2	Report
Week	13-Jul	O: 13.07							
commencing	20-Jul		O: 20.07						
Monday:	27-Jul								
	03-Aug	D: 03.08							
	10-Aug			O: 10.08					
	17-Aug		D: 17.08		O: 14.08 C: 21.08				O 17.08
	<i>Mid-Semester Study Period</i>			D:31.08					
	07-Sep					O: 07.09			
	14-Sep								
	21-Sep						O: 21.09		
	28-Sep								
	05-Oct					D: 05.10		O: 02.10 C: 09.10	
	12-Oct						D: 19.10		D: 19.10
		O=opens (available from); D=due (without late penalty); C=closes (not available after)							

Report Groups

- You should now be in active communication with your group mates
- Start / check contact via the In-Group Chat forum (Keeping In Touch section).
- Any non-responding students: contact me via Group Activity Feedback tool

DIALOGUE In-Group Chat

Use this to initiate contact with your group mate(s). Click Create to start a new chat, search for their name (see [Group List](#)), and select the appropriate student name; you can add more than one student to the chat if you are part of a group of 3.

{Unfortunately almost all student names can be selected here, since you are all part of the Auckland Internal student group, or the Distance student group. Please only initiate a chat with students in your group here.}

In the example: Auckland Internal student Andrew enters 'and' and finds all *other* Auckland Internal students with that combination of letters in either of their names. Having selected group mate Alexander, he can then enter 'Char' to find the third group member Charlie. {NB: these are not students in this class, so don't try to find them!}

▼ Open with

Users

No selection

▼ Open with

Users

✕ Alexander Barrington-Nash

(NB: please do not select students outside your group - this is not what this chat is designed for)

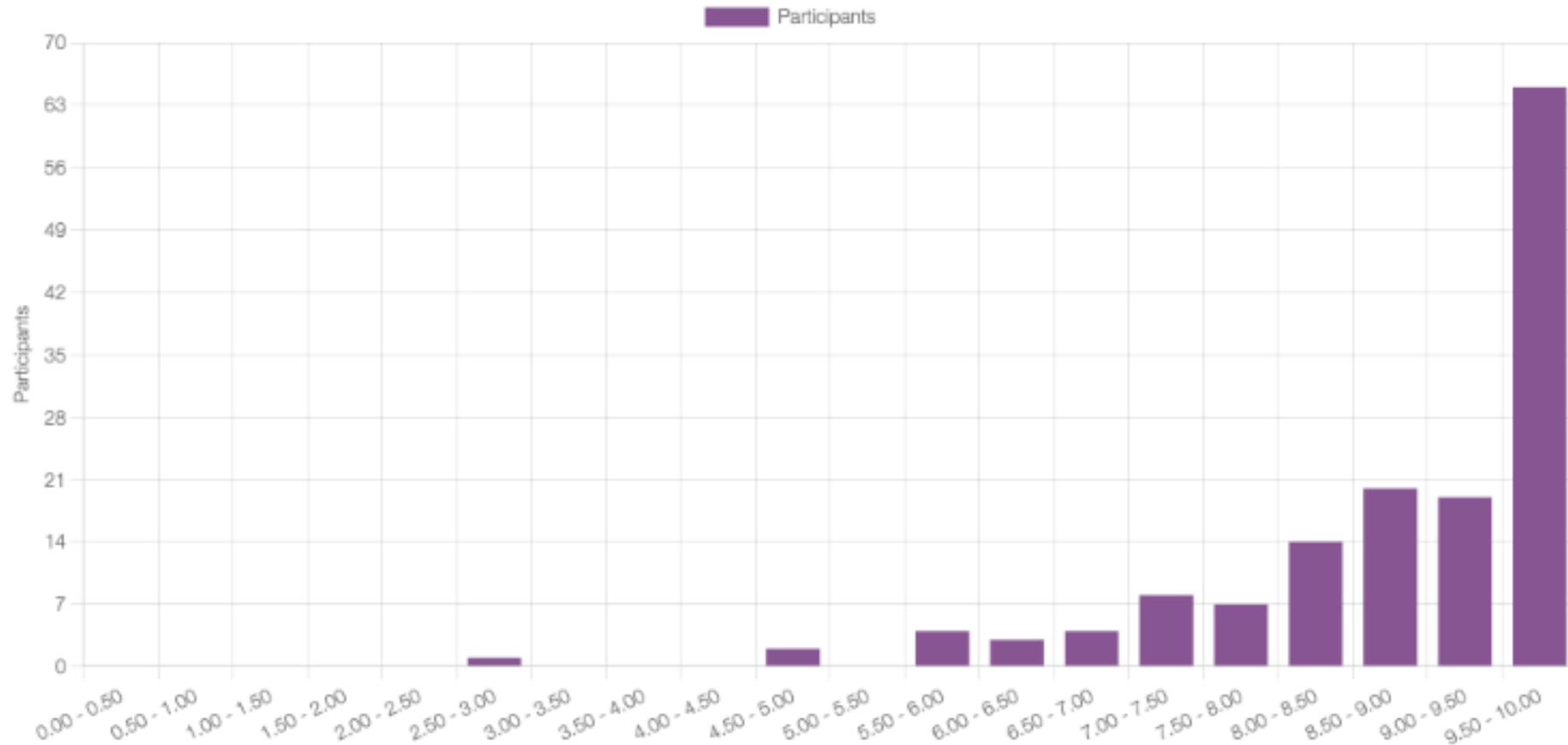


FEEDBACK

Group member activity (up to September 7)

Please indicate whether you have been in contact with your fellow group members, or would like to have your group reformed.

Test 1: marks



$\bar{x}$ of those attempting: 8.81 (2025: 8.26; 2024: 8.13, 2023: 7.64)

♂ $\bar{x}$: 8.74; ♀ $\bar{x}$: 8.86; Albany Internal $\bar{x}$: 8.92, Distance $\bar{x}$: 8.68

Assessments to Date

Group	Test 1	Labs 1&2	Labs 3&4	Total to date
Whole Class	8.81	8.69	8.10	23.29
Distance	8.68	8.93	8.28	23.07
Albany Internal	8.92	8.47	7.95	23.49
Female ♀	8.86	8.96	8.68	24.36
Male ♂	8.74	8.50	7.65	22.49

Problem:

How do we know our machines are working correctly?

Example: 5 Bean Canning Machines: target weight = 500g
Samples from each machine: (N=1000)

$$\bar{x} = 502 \quad 498 \quad 502 \quad 505 \quad 495$$

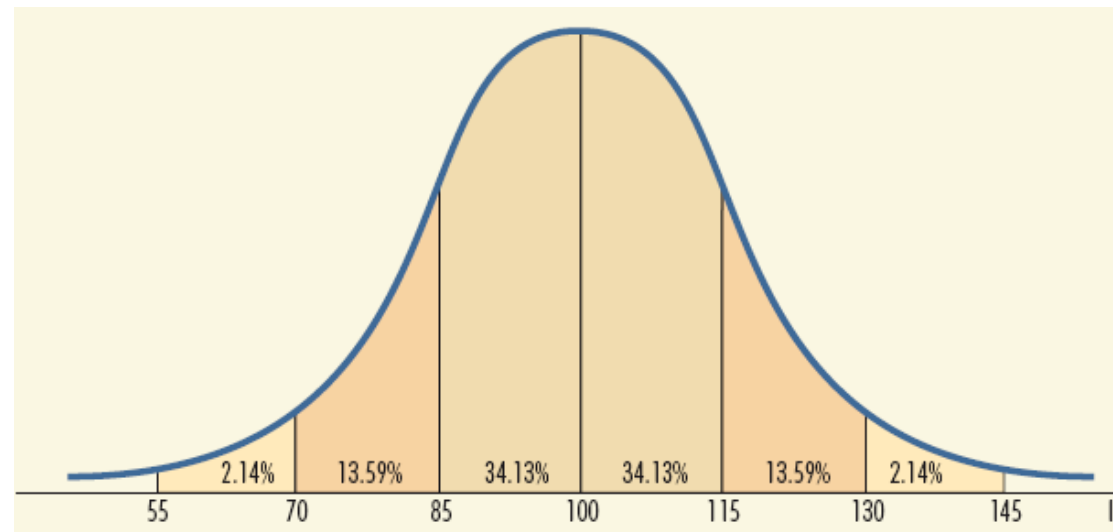
$$s = 2.5 \quad 5 \quad 10 \quad 10 \quad 15$$



1. Establish Null and Alternative Hypotheses
 H_o : machines are accurate; H_a : machines are inaccurate
 $H_o : \mu = 500$ $H_a : \mu \neq 500$
2. What test do we use?
For populations: Z test (or very large samples)
For samples: t-test
3. Z-test uses the Normal Distribution to determine the *critical value*

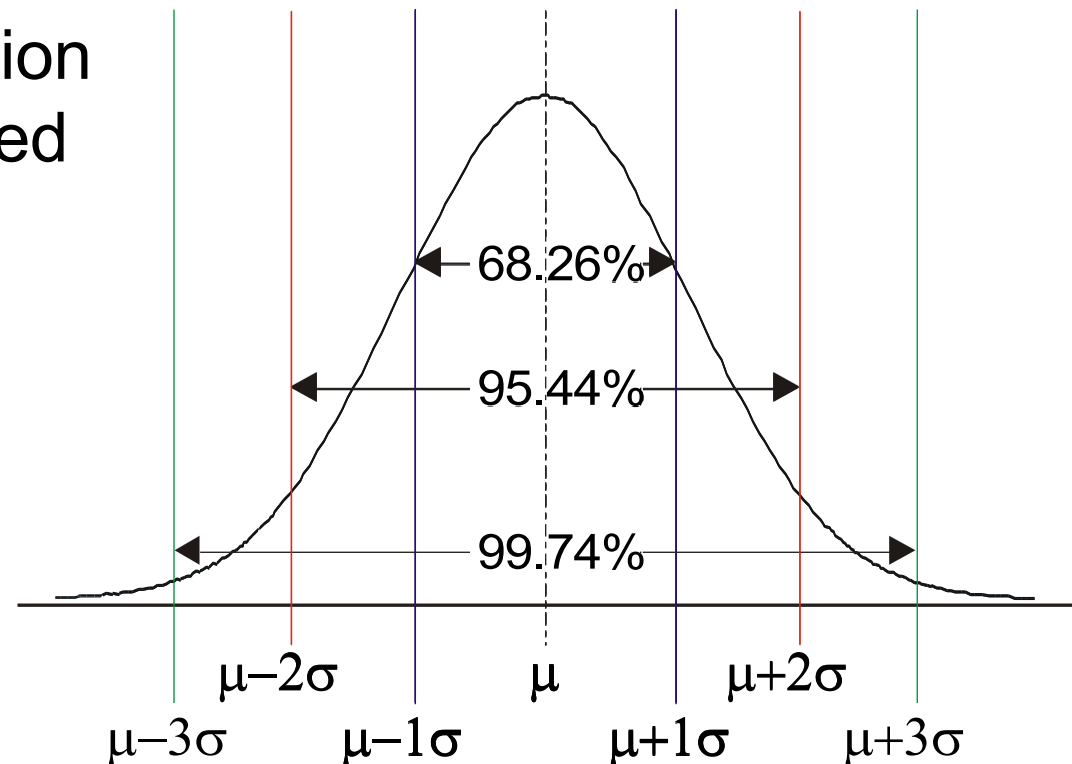
The Normal Distribution

- A symmetrical, bell-shaped distribution that describes the expected probability distribution of many chance occurrences
- With a normal bell-shaped curve, 99 per cent of its values are within ± 3 standard deviations from its mean, as the bell curve on IQ shows below.



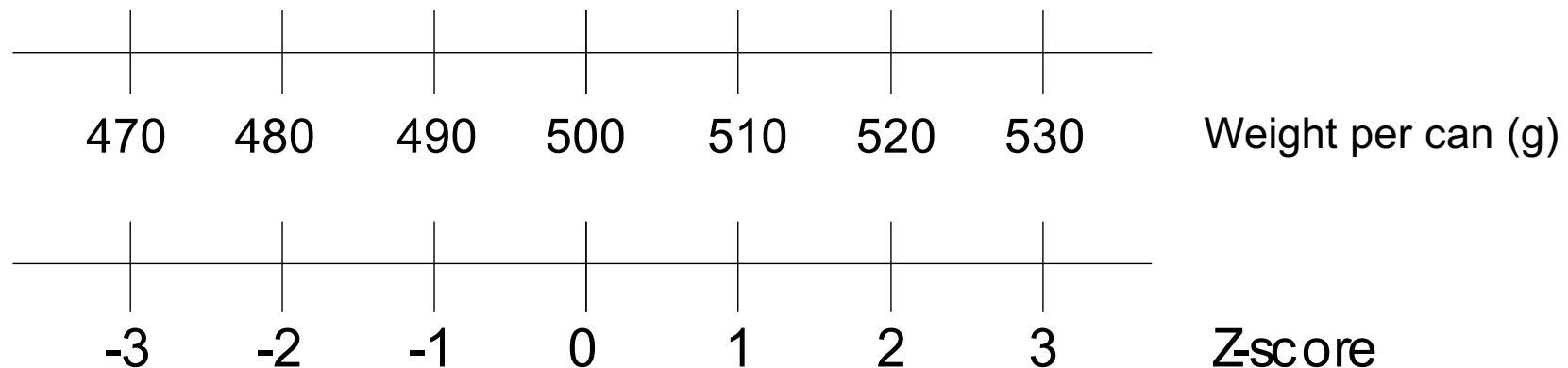
Normal Distribution: Properties

- 68.3% of observations fall within ± 1 standard deviation $[\sigma]$ of a normally distributed population mean $[\mu]$ (ie, Probability=0.683)
- 95.4% fall within $\pm 2\sigma$ (P=0.954)
- 99.7% fall within $\pm 3\sigma$ (P=0.997)



Z Scores: Standardised Data

- Standardised variable: directly comparable
- Formula for observations:
$$Z = \frac{x - \mu}{\sigma}$$
- Eg: 'true' machine: $\mu = 500\text{g}$ with $\sigma = 10\text{g}$

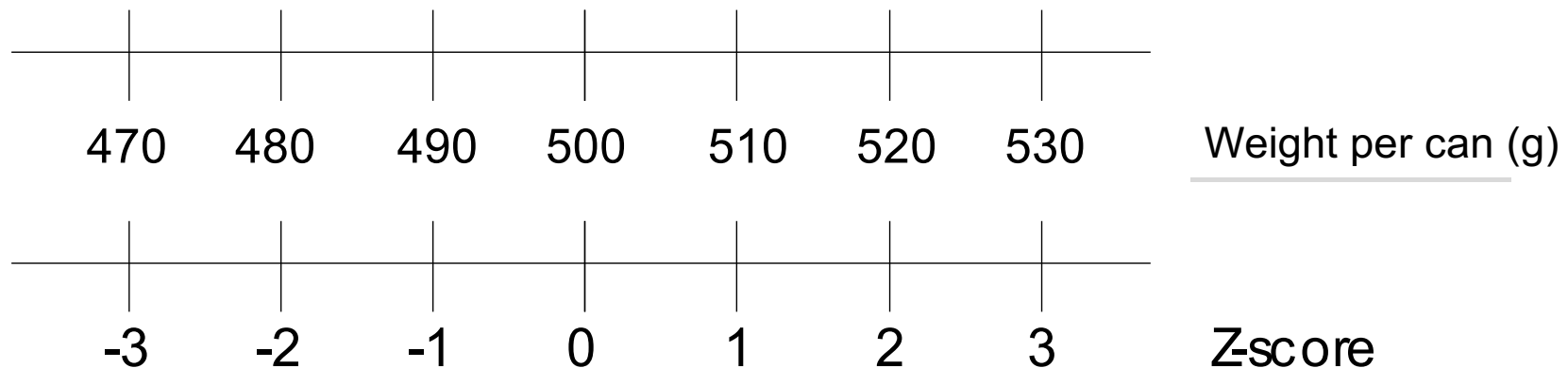


- Converted to mean of 0 and st dev of 1 (true for ANY normal distribution)

Z Scores: Standardised Data

- Formula for observations: $Z = \frac{x - \mu}{\sigma}$

- Eg: 'true' machine: $\mu = 500\text{g}$ with $\sigma = 10\text{g}$



eg1: 520g: $Z = \frac{520-500}{10} = 2$ standard deviations from mean

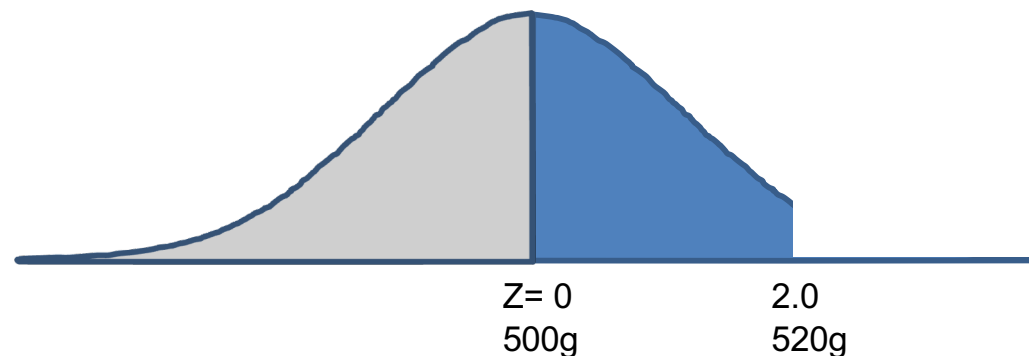
eg2: 475g: $Z = \frac{475-500}{10} = -2.5$ standard deviations from mean

Standardised Normal Table

Z standard deviations from the mean (units)	Z standard deviations from the mean (tenths of units)									
	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9
0.0	0.000	0.040	0.080	0.118	0.155	0.192	0.226	0.258	0.288	0.315
1.0	0.341	0.364	0.385	0.403	0.419	0.433	0.445	0.455	0.464	0.471
2.0	0.477	0.482	0.486	0.489	0.492	0.494	0.495	0.496	0.497	0.498
3.0	0.499	0.499	0.499	0.499	0.499	0.499	0.499	0.499	0.499	0.499

*Area under the segment of the normal curve extending (in one direction) from the mean to the point indicated by each row-column combination. For example, the table shows that about 68 per cent of normally distributed events can be expected to fall within 1.0 standard deviation on either side of the mean (0.341 times 2). An interval of almost 2.0 standard deviations around the mean will include 95 per cent of all cases.

- When $Z = 0$, the area under the curve = 0.477 (This means a 47.7% chance a can will weigh between 500g and 520g).



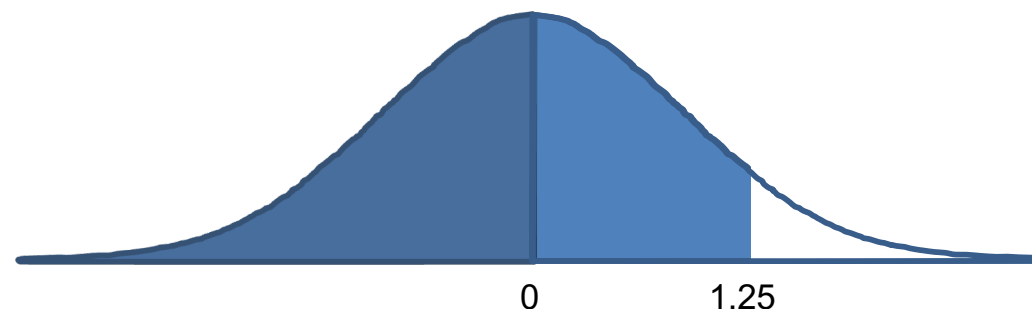
- When $Z = -2.5$, the area under the curve = 0.494 (ie a 49.4% chance of seeing a can between 475 and 500g)
- And a probability of 0.006 (0.6%) of a can being lighter than 475g

Standardising Values

- A toy manufacturer has mean sales of 9000 units and a standard deviation of 500 units. (Assume this is the population.)
- The company wishes to know whether wholesalers will demand between 7500 and 9625 units.

$$Z = \frac{X - \mu}{\sigma} = \frac{7500 - 9000}{500} = -3.00$$

$$Z = \frac{X - \mu}{\sigma} = \frac{9625 - 9000}{500} = 1.25$$



Standardised Normal Table

Z standard deviations from the mean (units)	Z standard deviations from the mean (tenths of units)									
	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9
0.0	0.000	0.040	0.080	0.118	0.155	0.192	0.226	0.258	0.288	0.315
1.0	0.341	0.364	0.385	0.403	0.419	0.433	0.445	0.455	0.464	0.471
2.0	0.477	0.482	0.486	0.489	0.492	0.494	0.495	0.496	0.497	0.498
3.0	0.499	0.499	0.499	0.499	0.499	0.499	0.499	0.499	0.499	0.499

*Area under the segment of the normal curve extending (in one direction) from the mean to the point indicated by each row-column combination. For example, the table shows that about 68 per cent of normally distributed events can be expected to fall within 1.0 standard deviation on either side of the mean (0.341 times 2). An interval of almost 2.0 standard deviations around the mean will include 95 per cent of all cases.

- When $Z = -3.00$, the area under the curve = 0.499.
- When $Z =$, the area under the curve = (average of)
- The total area under the curve = $0.499 +$
= .
- There is a probability that sales will fall in that range.

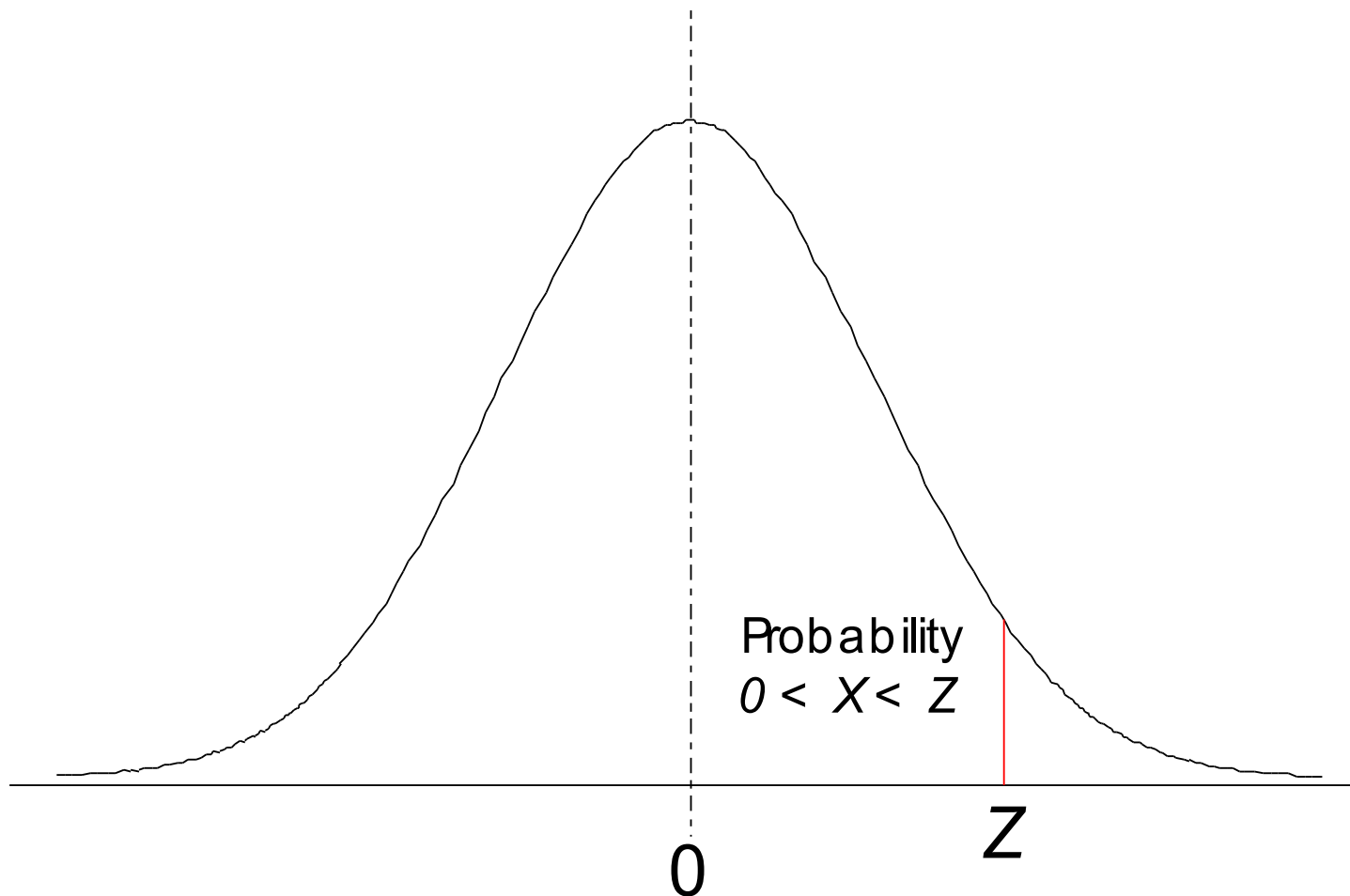
Z Table – key numbers

Standard deviations (<i>z values / scores</i>)	Probability $-z < P < z$	Probability $P < -z$ or $P > z$
± 1	0.6826	0.3174
± 1.645	0.9	0.10
± 1.96	0.95	0.05
± 2	0.9544	0.0456
± 2.576 (2.58)	0.99	0.01
± 3	0.9974	0.0026

Calculating Z

$$Z = \frac{X - \mu}{\sigma}$$

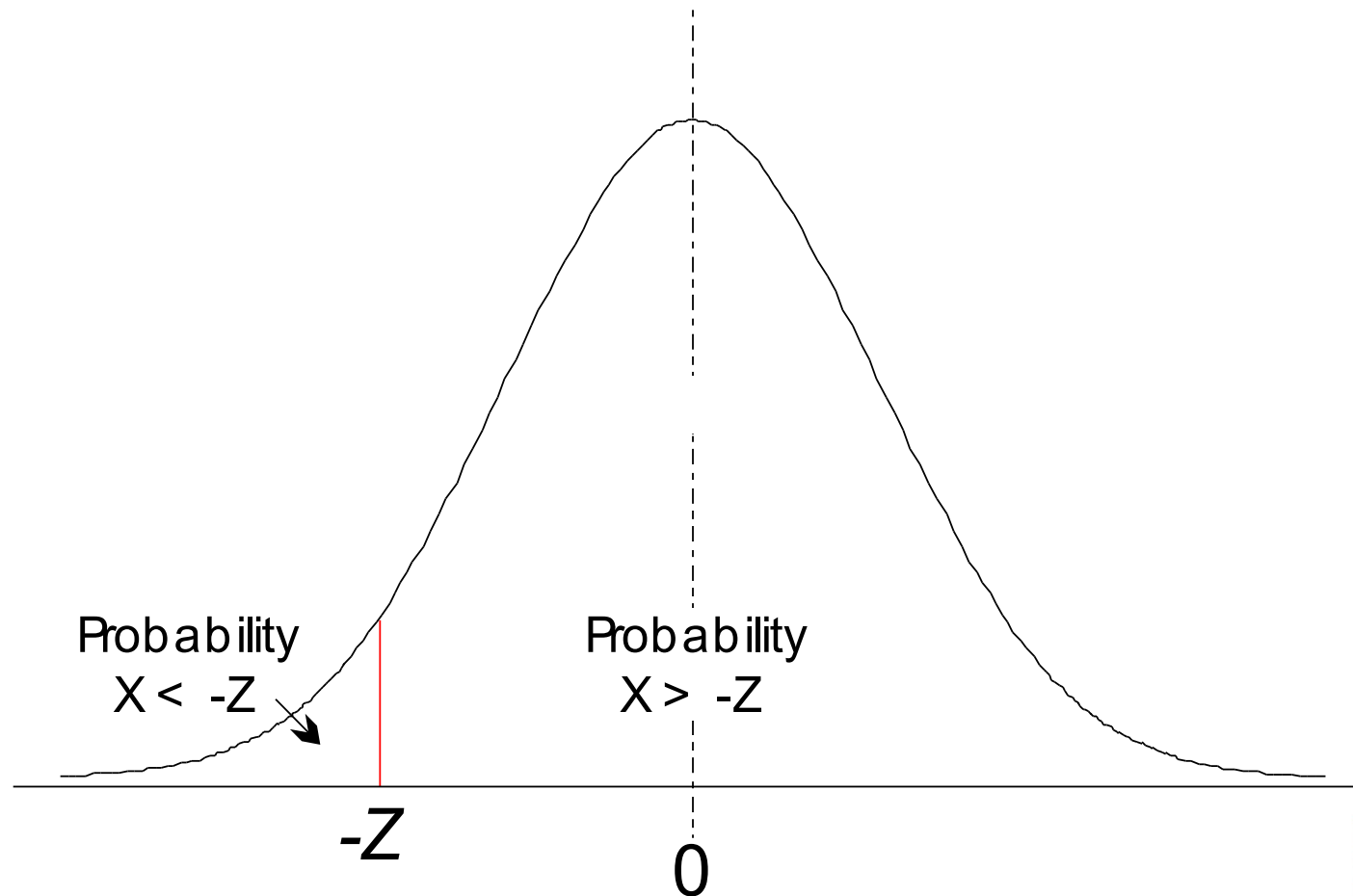
ZA: Centred



Calculating Z: One-tailed

~~Z~~: One-tailed

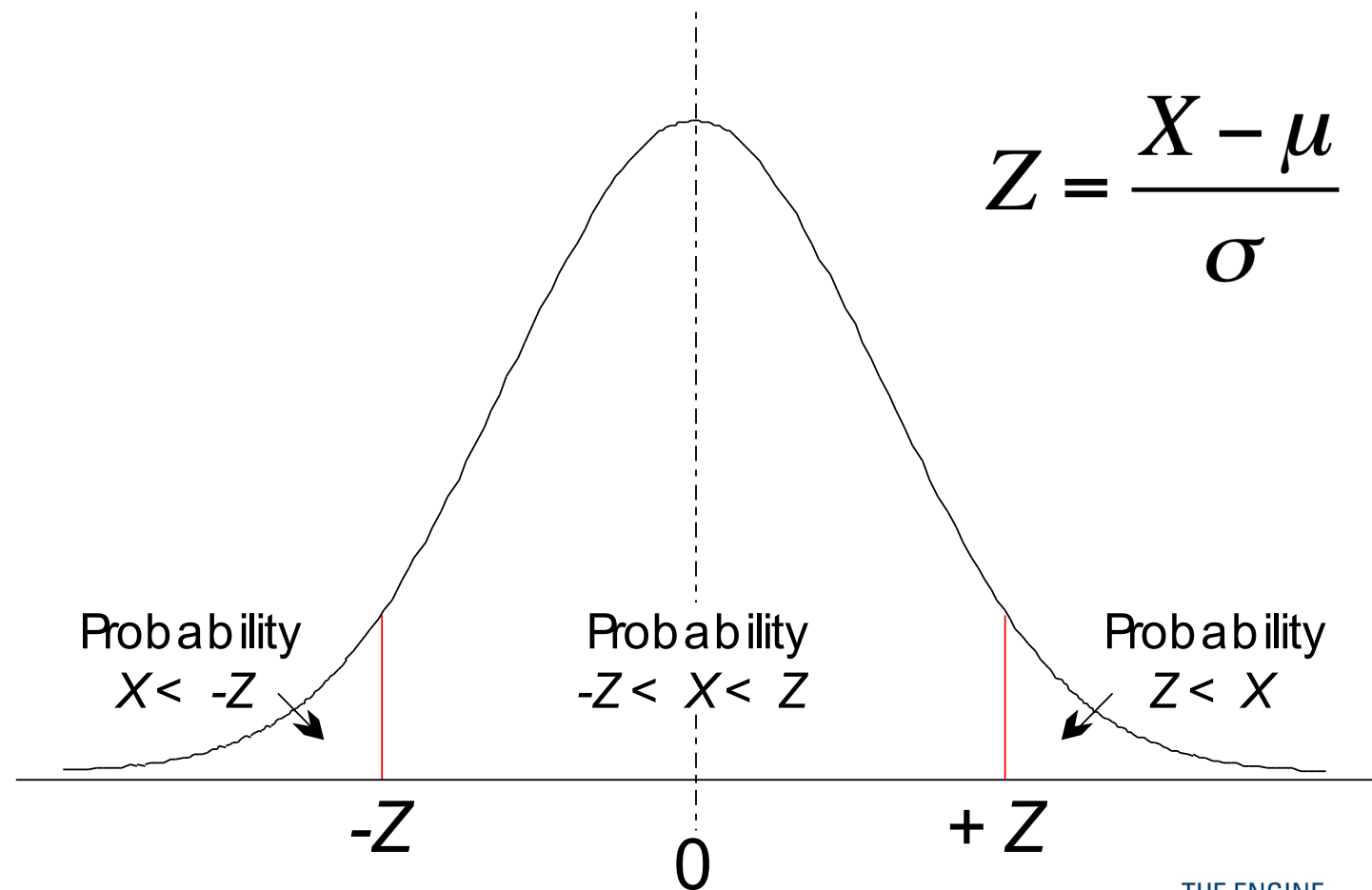
$$Z = \frac{X - \mu}{\sigma}$$



Calculating Z: Two-tailed

ZC: Two-tailed

$$Z = \frac{X - \mu}{\sigma}$$



Observations, Samples and Populations

- We can standardise individual observations using the formula

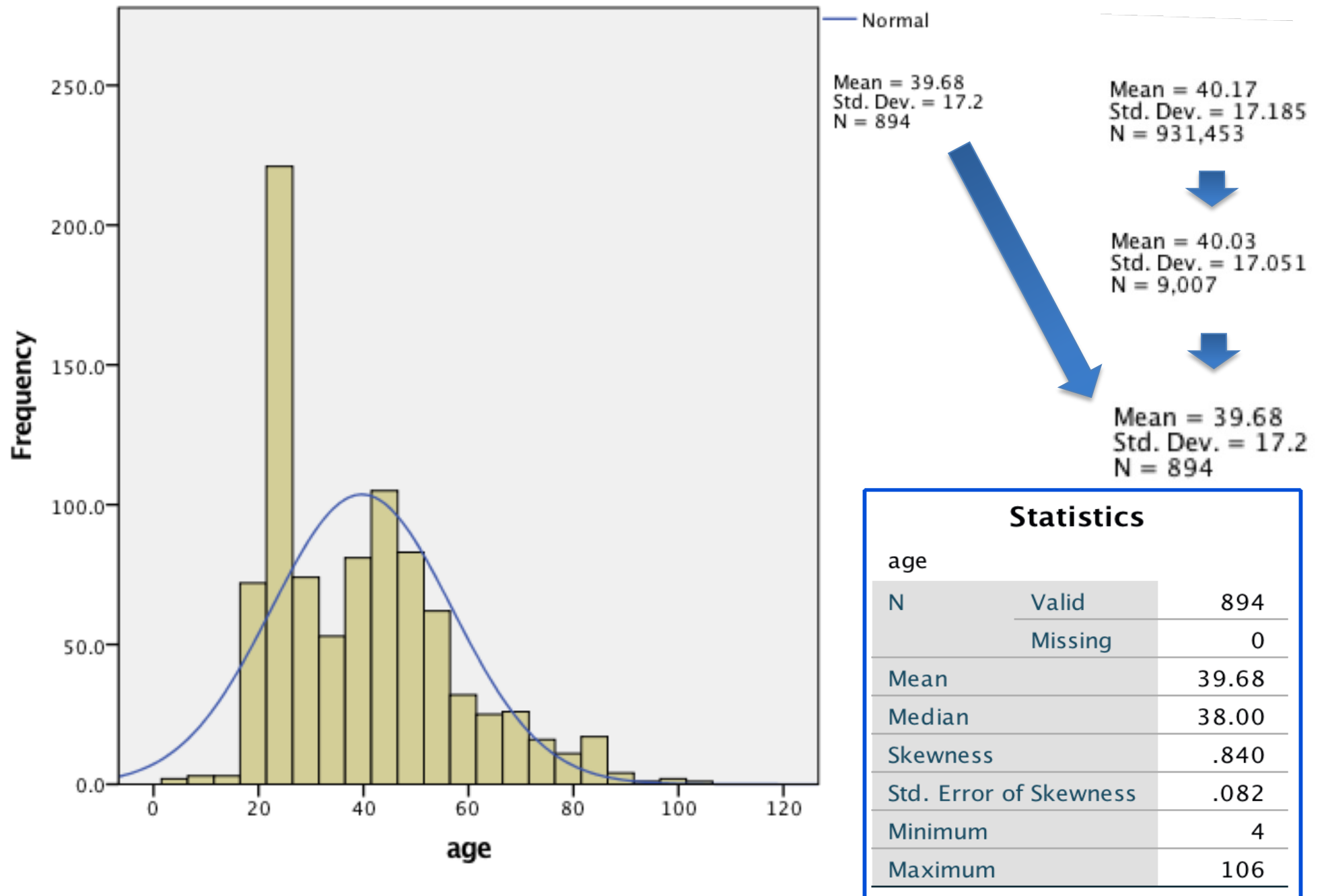
$$Z = \frac{X - \mu}{\sigma}$$

- What about samples themselves? What can we *infer* about the population from which they were drawn?
- This is the branch of statistics known as *inferential statistics*
- Its key theorem is the Central Limit Theorem, which relates samples to populations

Population, sample and sampling distribution

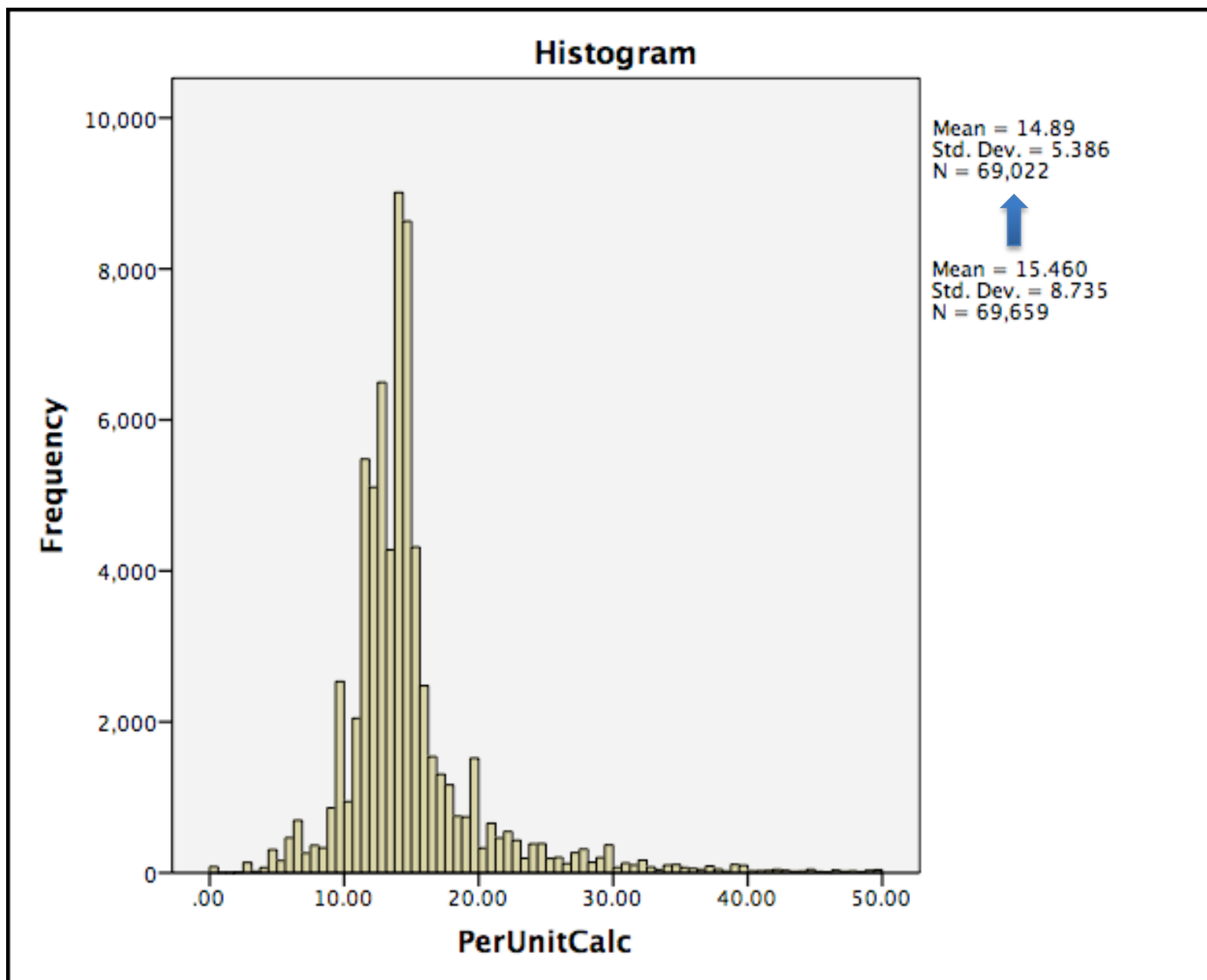
- **Population distribution:** A frequency distribution of the elements of a population
- **Sample distribution:** A frequency distribution of a sample
- **Sampling distribution:** A theoretical probability of sample means for all possible samples of a certain size drawn from a particular population
 - Addresses the question ‘What would happen if we were to draw a large number of samples, each having n elements, from a specified population?’
- **Standard error of the mean:** The standard error of the sampling distribution

Santander: 0.1% Sample (N=894) (shown T5)



CDNow: Full Sample (N=69,659)

CD sales transactions from Jan 1997 – June 1998



CDNow : 10 ~1% Samples

		Mean	St Dev	Count	Max
Sample statistics	1 $\bar{x}$	15.21	s 6.885	n 650	100
	2	15.87	10.176	725	167
	3	14.98	6.668	642	81
	4	15.15	6.605	697	87
	5	15.85	12.509	687	284
	6	15.66	8.831	656	124
	7	15.27	8.528	691	128
	8	15.04	8.555	643	137
	9	15.47	9.181	673	163
	10	15.43	7.776	690	146
		15.39*	0.39**	10***	
Population	μ	15.46	σ 8.735	69659	

* Mean of sample means

** Standard deviation of sample means
("sample standard error")

*** number of samples drawn

Population standard error: $SE_{\mu} = \frac{\sigma}{\sqrt{n}} = \frac{8.735}{\sqrt{69659}} = \frac{8.735}{263.93} = 0.033$

Central Limit Theorem

For samples of 30 or larger (“large” sample):

- the set of sample means will be normally distributed (even if one sample, or the population, isn’t, as long as the samples are randomly selected)

- the mean of the sample means is a good (unbiased) approximation of the population mean μ

- the standard deviation of the sample means (the “standard error of the sample mean”) is smaller than the population standard deviation – if samples large enough, will equal

$$S_{\bar{x}}$$

$$SE_{\mu} = \frac{\sigma}{\sqrt{n}}$$

- We can estimate standard error from samples: (larger samples will have a smaller standard error)

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

CDNow : 10 ~1% Samples

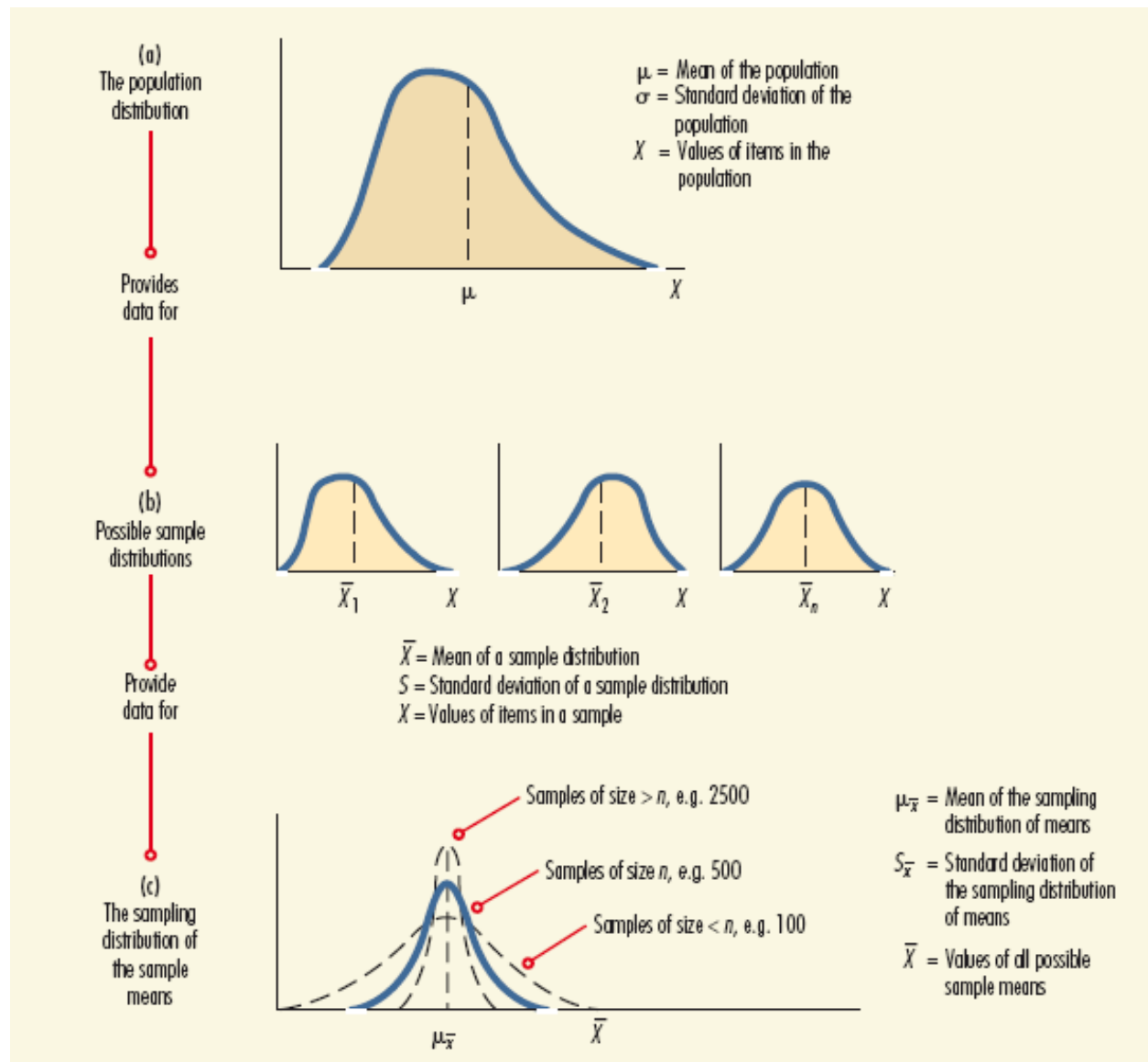
		Mean	St Dev	Count	Std Error
Sample statistics	1	$\bar{x}$ 15.21	s 6.885	n 650	0.270
	2	15.87	10.176	725	0.378
	3	14.98	6.668	642	0.263
	4	15.15	6.605	697	0.250
	5	15.85	12.509	687	0.477
	6	15.66	8.831	656	0.345
	7	15.27	8.528	691	0.324
	8	15.04	8.555	643	0.337
	9	15.47	9.181	673	0.354
	10	15.43	7.776	690	0.296
		15.39	0.39	10	
Population	μ	15.46	σ 8.735	69659	0.033
'Large' Sample		15.33	7.865	6776	0.096

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

Outlier: x=\$284

$$SE_{\mu} = \frac{\sigma}{\sqrt{n}}$$

Population, sample and sampling distribution



Confidence intervals

- A **confidence interval estimate** is based on the knowledge that the population mean is the sample mean plus or minus a small sampling error.
 - After calculating an interval estimate, we can determine how probable it is that the population mean will fall within this range of statistical values.
- **Confidence level** is a percentage that indicates the long-run probability that the results will be correct.

Confidence Intervals and Estimation

- Sample mean is an estimate of the *true* population mean
 $\bar{x}$ for samples, μ for population
- We can create an *interval*, based on the standard error of the sample, within which we predict the population mean will lie with a given *level of confidence* (eg: 95% - we would be right “19 times out of 20”)
- The interval has a lower confidence limit (LCL) and an upper confidence limit (UCL)
- $LCL = \text{mean } (\bar{x}) \text{ minus } [\text{standard error (SE)} \times z \text{ value (associated with the level of confidence)}]$
- Eg: for 95% CI:
$$LCL = \bar{x} - 1.96SE_{\bar{x}} \quad \text{or} \quad U / LCL = \bar{x} \pm 1.96 \frac{s}{\sqrt{n}}$$
$$UCL = \bar{x} + 1.96SE_{\bar{x}}$$

Confidence Intervals: Estimation

Confidence levels are expressed as percentages: we can be “95% confident” that the true population mean falls between the LCL and UCL determined by the sample mean, standard error and z value.

If the underlying distribution is **normal**, then we can say:

- there is a probability of 0.683 that $(\bar{x} - SE_{\bar{x}}) \leq \mu \leq (\bar{x} + SE_{\bar{x}})$
- there is a probability of 0.954 that $(\bar{x} - 2SE_{\bar{x}}) \leq \mu \leq (\bar{x} + 2SE_{\bar{x}})$
- there is a probability of 0.997 that $(\bar{x} - 3SE_{\bar{x}}) \leq \mu \leq (\bar{x} + 3SE_{\bar{x}})$

Or more typically:

- a probability of 0.95 that $(\bar{x} - 1.96SE_{\bar{x}}) \leq \mu \leq (\bar{x} + 1.96SE_{\bar{x}})$

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

Confidence Intervals: Example

- A sporting goods store caters to working women who golf; a sample of 100 customers gives a mean age of 37.5 years and standard deviation of 12.0 years.

$$\bar{x} = 37.5 \text{ years}$$

$$S = 12.0 \text{ years}$$

$$n = 100$$

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}} = \frac{12}{\sqrt{100}} = 1.2$$

- Suppose the store wishes to be 95 per cent confident of the population mean age (ie, that the sample estimate range will include the population parameter).

Confidence Intervals: Example

- Including 95 per cent of the area requires that 47.5 per cent of the distribution on each side be included.
- Referring to Z table (Table 6, page 784), we find that 0.475 corresponds to the Z-value 1.96.

$$(\bar{x} - 1.96SE_{\bar{x}}) \leq \mu \leq (\bar{x} + 1.96SE_{\bar{x}})$$

- 95 per cent of the time μ is in range of years.
- Larger sample sizes give greater confidence
 - ➔ larger N = larger denominator for standard error
 - ➔ thus smaller standard error
 - ➔ thus narrower confidence interval

$$\frac{s}{\sqrt{n}}$$

CDNow : 10 ~1% Samples

		Mean	St Dev	Count	Std Error
Sample statistics	1 $\bar{x}$	15.21	s 6.885	n 650	0.270
	2	15.87	10.176	725	0.378
	3	14.98	6.668	642	0.263
	4	15.15	6.605	697	0.250
	5	15.85	12.509	687	0.477
	6	15.66	8.831	656	0.345
	7	15.27	8.528	691	0.324
	8	15.04	8.555	643	0.337
	9	15.47	9.181	673	0.354
	10	15.43	7.776	690	0.296
		15.39	0.39	10	
Population	μ	15.46	σ 8.735	69659	0.033
'Large' Sample		15.33	7.865	6776	0.096

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

Outlier: x=\$284

$$SE_{\mu} = \frac{\sigma}{\sqrt{n}}$$

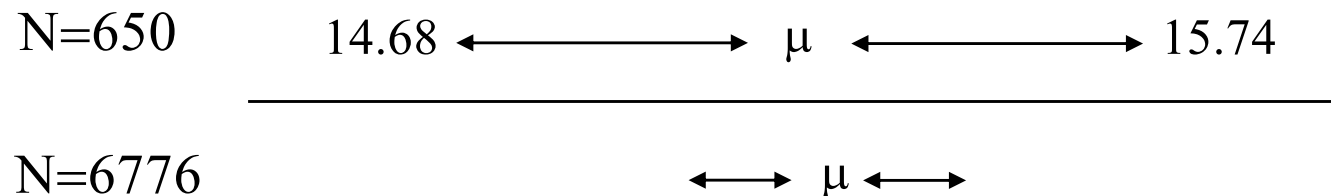
Confidence Intervals: CD spend

- Larger sample sizes give greater confidence about “true” value of μ
- “Small” sample (650): “Large” sample (6776):

$$= 15.21 \pm 1.96 \frac{6.885}{\sqrt{650}}$$

$$= 15.21 \pm 0.53$$

$$= 14.68 - -15.74$$

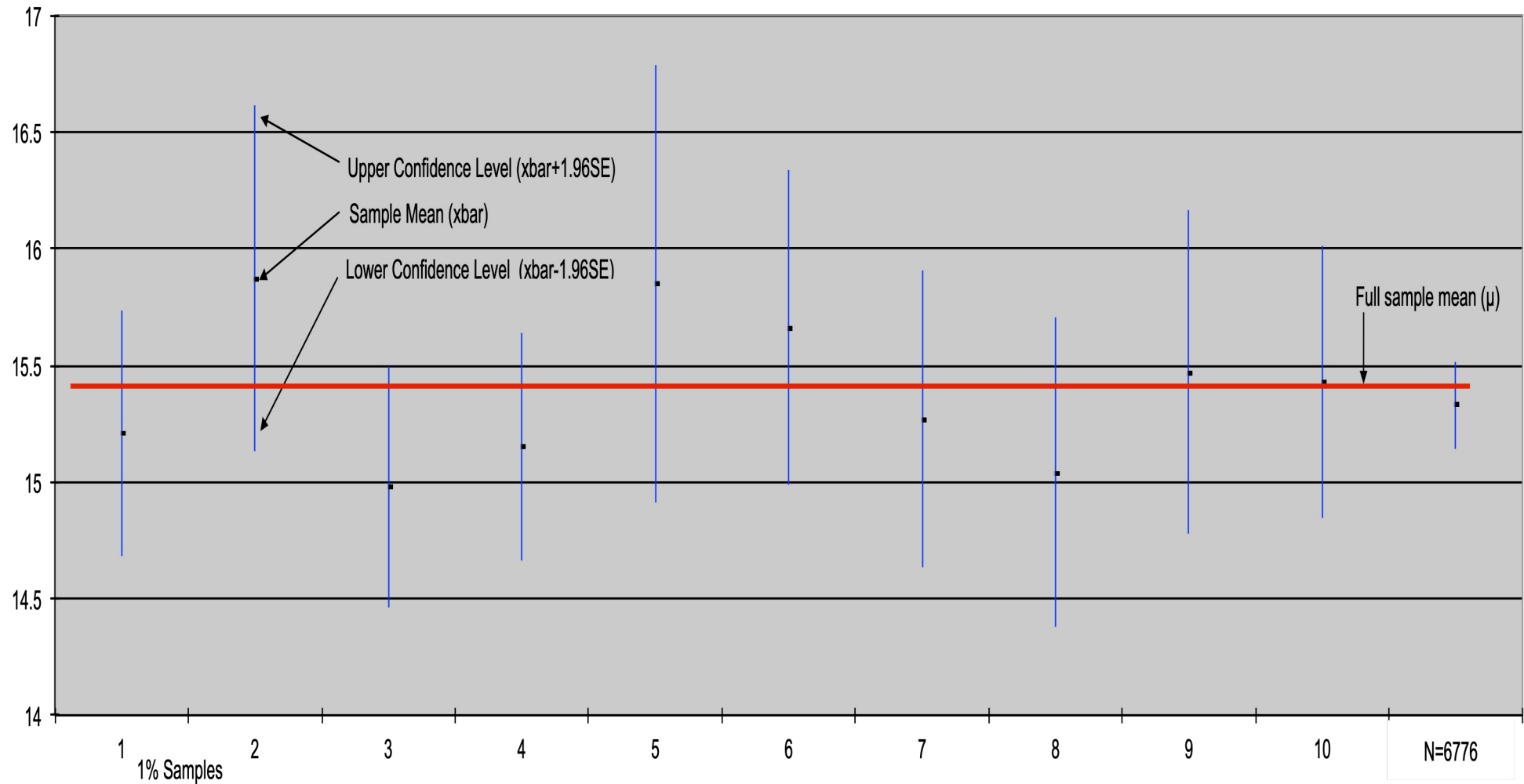


CDNow : 10 ~1% Samples

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

		Mean	St Dev	Count	LCL	UCL
Sample statistics	1 $\bar{x}$	15.21	s 6.885	n 650	14.68	15.74
	2	15.87	10.176	725	15.13	16.61
	3	14.98	6.668	642	14.46	15.50
	4	15.15	6.605	697	14.66	15.64
	5	15.85	12.509	687	14.91	16.79
	6	15.66	8.831	656	14.98	16.34
	7	15.27	8.528	691	14.63	15.91
	8	15.04	8.555	643	14.38	15.70
	9	15.47	9.181	673	14.78	16.16
	10	15.43	7.776	690	14.85	16.01
Population		μ 15.46		69659		
'Large' Sample		15.33	7.865	6776		

Confidence Intervals: CDNow



Hypothesis Testing: Means

- A **hypothesis** is an unproven proposition or supposition that tentatively explains certain facts or phenomena.
 - Empirically testable
- A **null hypothesis** is a statement about a status quo.
 - Any change from what has been thought to be true will be due entirely to random sampling error.
- An **alternative hypothesis** is a statement indicating the opposite of the null hypothesis.

Hypothesis Testing: Means

- For example, frequent customers spend more in a store per shopping trip than infrequent customers
 - The **null** hypothesis is where there is **no difference** between frequent vs infrequent customers and their expenditure per shopping trip.
 - The **alternative hypothesis** occurs when there is a difference in expenditure between frequent and infrequent customers.

Hypothesis Testing: Means

- **Significance level** is the critical probability in choosing between the null and alternative hypotheses
 - Eg $\alpha = 0.05$: happy to accept the probability of being wrong 1 time in 20
- Have two-tailed and one-tailed tests:
 - Two-tailed: test is of difference – doesn't matter if bigger or smaller (both tails count)
 - One-tailed: direction matters (only one tail counts)

Two-Tailed 'Z' Test

Example: Bean Canning Machines: target weight = 500g

Machine samples (N=25) $\bar{x} = 502$ 498 502 505 495

$s = 2.5$ 5 10 10 15

$z =$

1. Establish Null and Alternative Hypotheses
 H_o : machines are accurate; H_a : machines are inaccurate

$$H_o : \mu = 500 \qquad H_a : \mu \neq 500$$

2. Decide critical level of significance:

$\alpha = 0.05$ (ie, 2.5% in each tail) $\therefore z_c = 1.96$

3. Calculate "test statistic"

$$Z = \frac{x - \bar{x}}{\frac{s}{\sqrt{n}}} \quad Z = \frac{502 - 500}{\frac{2.5}{\sqrt{25}}} = \frac{2}{\frac{2.5}{5}} = 4.0$$

4. Compare test statistic against critical level of significance
Terminology: $\alpha = 0.05$ (ie, 2.5% in each tail) $\therefore z_c = 1.96$

5. Decision: if test statistic $> z_c$ reject H_o ,
otherwise "can't reject H_o "

Z-Table: Two-tailed (p. 784)

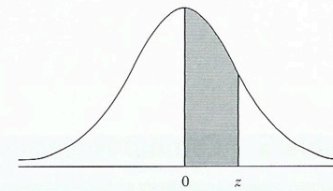


Table 6 Areas under the standard normal curve

This table gives the area between 0 and +z

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09	z
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359	0.0
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753	0.1
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141	0.2
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517	0.3
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879	0.4
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224	0.5
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549	0.6
0.7	.2580	.2611	.2642	.2673	.2703	.2734	.2764	.2793	.2823	.2852	0.7
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133	0.8
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389	0.9
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621	1.0
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830	1.1
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015	1.2
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177	1.3
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4192	.4306	.4319	1.4
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441	1.5
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545	1.6
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633	1.7
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706	1.8
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767	1.9
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817	2.0
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857	2.1
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890	2.2
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916	2.3
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936	2.4
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952	2.5
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964	2.6
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974	2.7
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981	2.8
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986	2.9
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990	3.0
3.1	.4990	.4991	.4991	.4991	.4992	.4992	.4992	.4992	.4993	.4993	3.1
3.2	.4993	.4993	.4994	.4994	.4994	.4994	.4994	.4995	.4995	.4995	3.2
3.3	.4995	.4995	.4995	.4996	.4996	.4996	.4996	.4996	.4996	.4997	3.3
3.4	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4998	3.4
3.5	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	3.5
3.6	.4998	.4998	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	3.6
3.7	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	3.7
3.8	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	3.8
3.9	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000	3.9

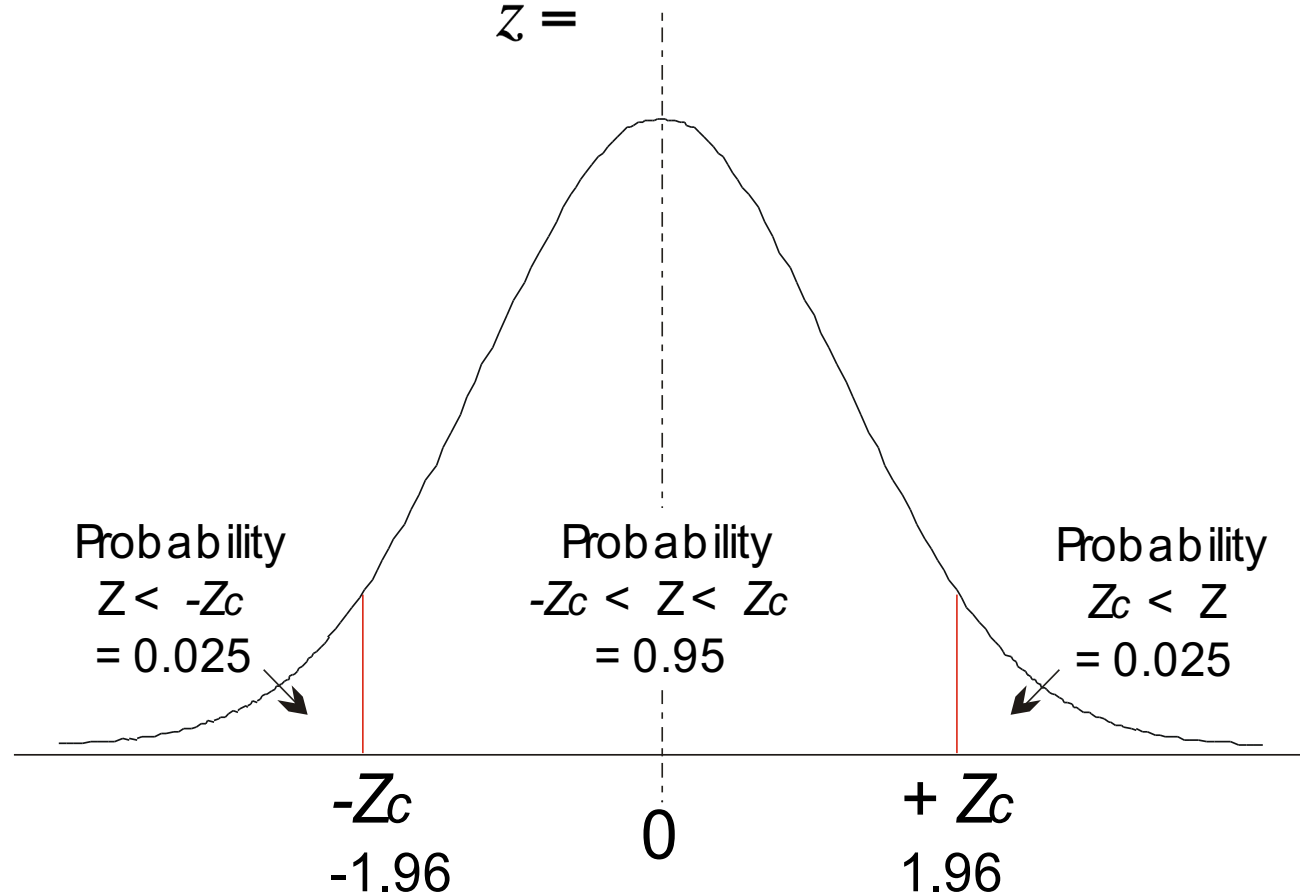
Calculating Z: Two-tailed

$$Z = \frac{x - \bar{x}}{\frac{s}{\sqrt{n}}}$$

$\bar{x} = 502 \quad 498 \quad 502 \quad 505 \quad 495$

$s = 2.5 \quad 5 \quad 10 \quad 10 \quad 15$

$z =$



One-tailed Z Test

Suspect sample mean is smaller than target

Example: Bean Canning Machines: target weight = 500g

Machine samples (N=25)

$$\bar{x} = 502 \quad 498 \quad 502 \quad 505 \quad 495$$

$$s = 2.5 \quad 5 \quad 10 \quad 10 \quad 15$$

$$z =$$

1. Establish Null and Alternative Hypotheses

H_0 = machines are accurate; H_a : machines are under-filling

$$H_0 : \mu = 500$$

$$H_a : \mu < 500$$

2. Decide critical level of significance:

$$\alpha = 0.05 \text{ (ie, 5\% in left tail)} \therefore z_c = -1.645$$

3. Calculate "test statistic"

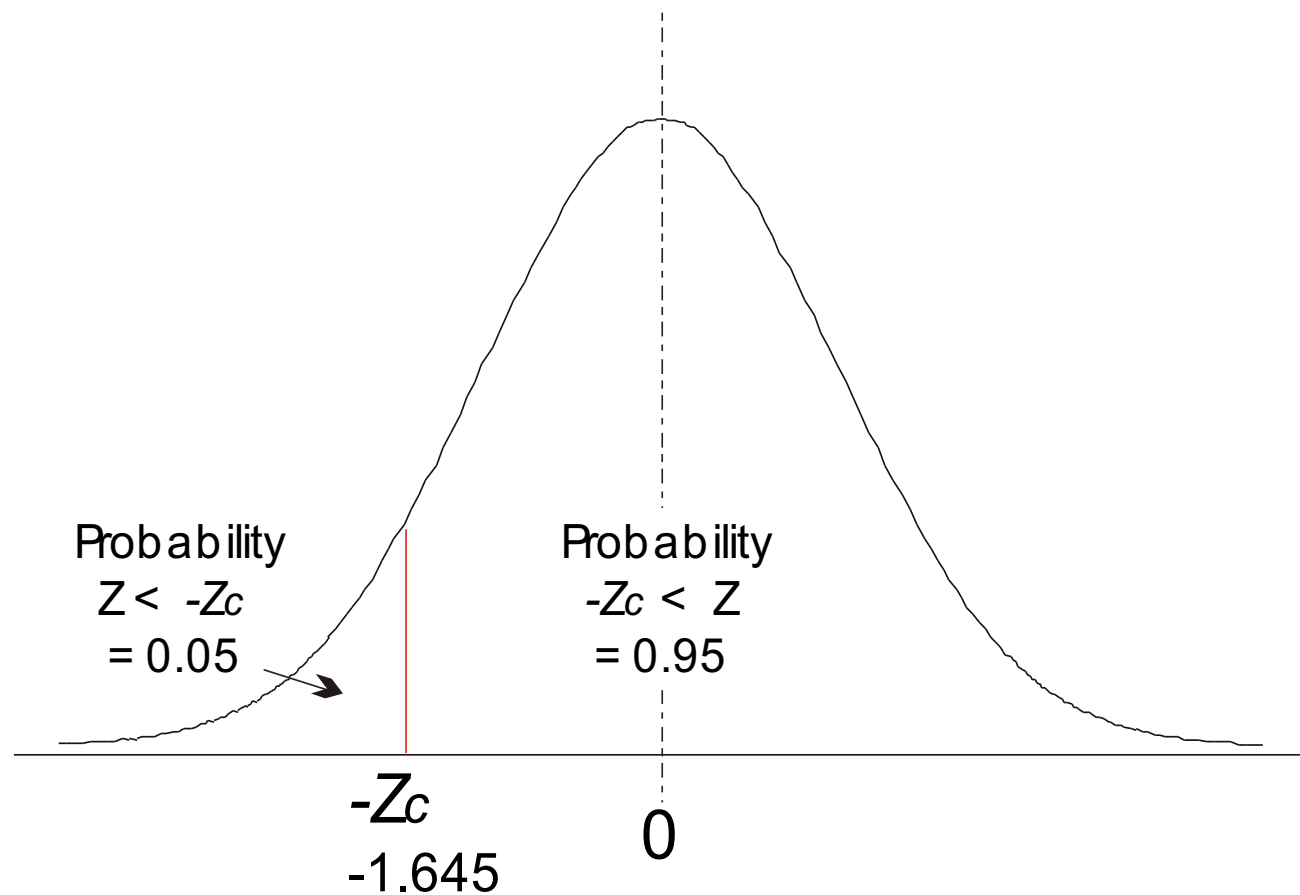
$$Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} \quad Z = \frac{502 - 500}{\frac{2.5}{\sqrt{25}}} = \frac{2}{\frac{2.5}{5}} = 4.0$$

4. Compare test statistic against critical level of significance

5. Decision: if test statistic left of $-z_c$ then reject H_0

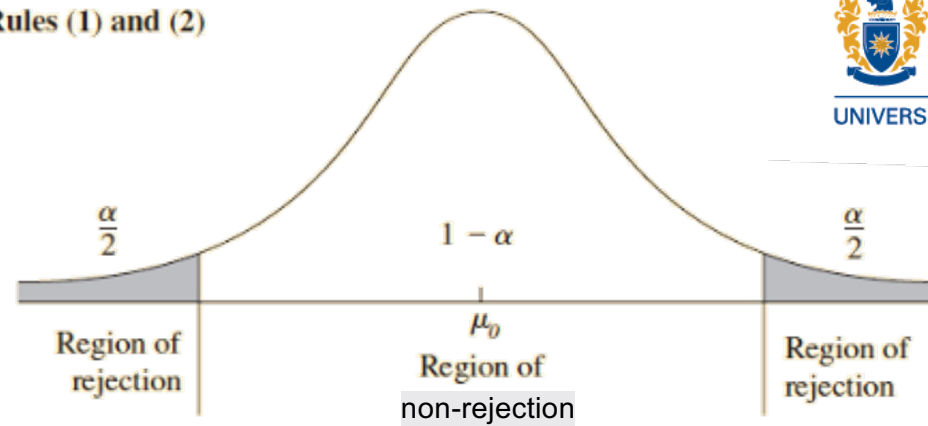
Calculating Z: One-tailed

$$Z = \frac{x - \bar{x}}{\frac{s}{\sqrt{n}}}$$

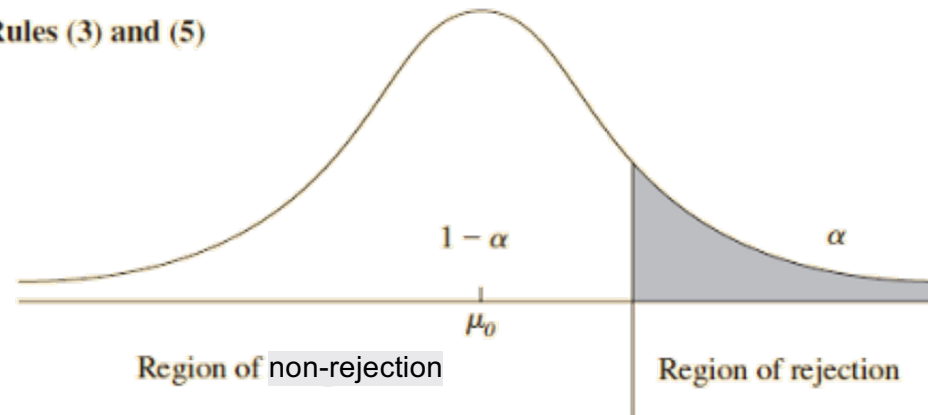


Outcomes

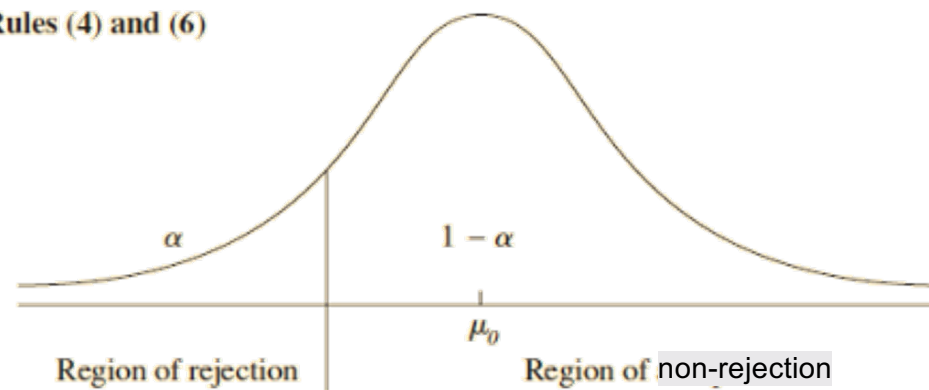
Rules (1) and (2)



Rules (3) and (5)



Rules (4) and (6)



Z-Table: One-tailed

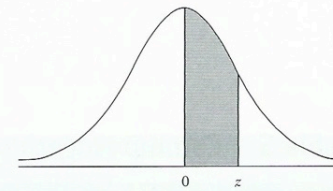


Table 6 Areas under the standard normal curve

This table gives the area between 0 and +z

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09	z
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359	0.0
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753	0.1
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141	0.2
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517	0.3
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879	0.4
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224	0.5
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549	0.6
0.7	.2580	.2611	.2642	.2673	.2703	.2734	.2764	.2793	.2823	.2852	0.7
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133	0.8
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389	0.9
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621	1.0
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830	1.1
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015	1.2
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177	1.3
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4192	.4306	.4319	1.4
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441	1.5
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545	1.6
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633	1.7
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706	1.8
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767	1.9
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817	2.0
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857	2.1
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890	2.2
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916	2.3
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936	2.4
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952	2.5
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964	2.6
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974	2.7
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981	2.8
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986	2.9
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990	3.0
3.1	.4990	.4991	.4991	.4991	.4992	.4992	.4992	.4992	.4993	.4993	3.1
3.2	.4993	.4993	.4994	.4994	.4994	.4994	.4994	.4995	.4995	.4995	3.2
3.3	.4995	.4995	.4995	.4996	.4996	.4996	.4996	.4996	.4996	.4997	3.3
3.4	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4998	3.4
3.5	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	3.5
3.6	.4998	.4998	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	3.6
3.7	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	3.7
3.8	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	3.8
3.9	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000	3.9

Multivariate Hypothesis Testing

1. Null Hypothesis: two samples are the same $H_o : \mu_1 = \mu_2$
 $H_o : \mu_1 - \mu_2 = 0$
- Alternative Hypothesis: are different (two-tail)
 $H_A : \mu_1 \neq \mu_2 \quad H_A = \mu_1 - \mu_2 \neq 0 \quad \bar{x}_1 = 502 \quad \bar{x}_2 = 498$
- or one is larger than the other (one-tail) $s_1 = 5 \quad s_2 = 10$

$H_A : \mu_1 > \mu_2 \quad H_A = \mu_1 - \mu_2 > 0$

 $n_1 = 40 \quad n_2 = 50$

2. Test Statistic: $Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$

$$Z = \frac{502 - 498}{\sqrt{\frac{5^2}{40} + \frac{10^2}{50}}} \quad Z = \frac{502 - 498}{\sqrt{\frac{25}{40} + \frac{100}{50}}}$$
3. Compare test statistic vs critical level of significance

$\alpha=0.05$ (5% in right tail) $\therefore z_c = 1.645$

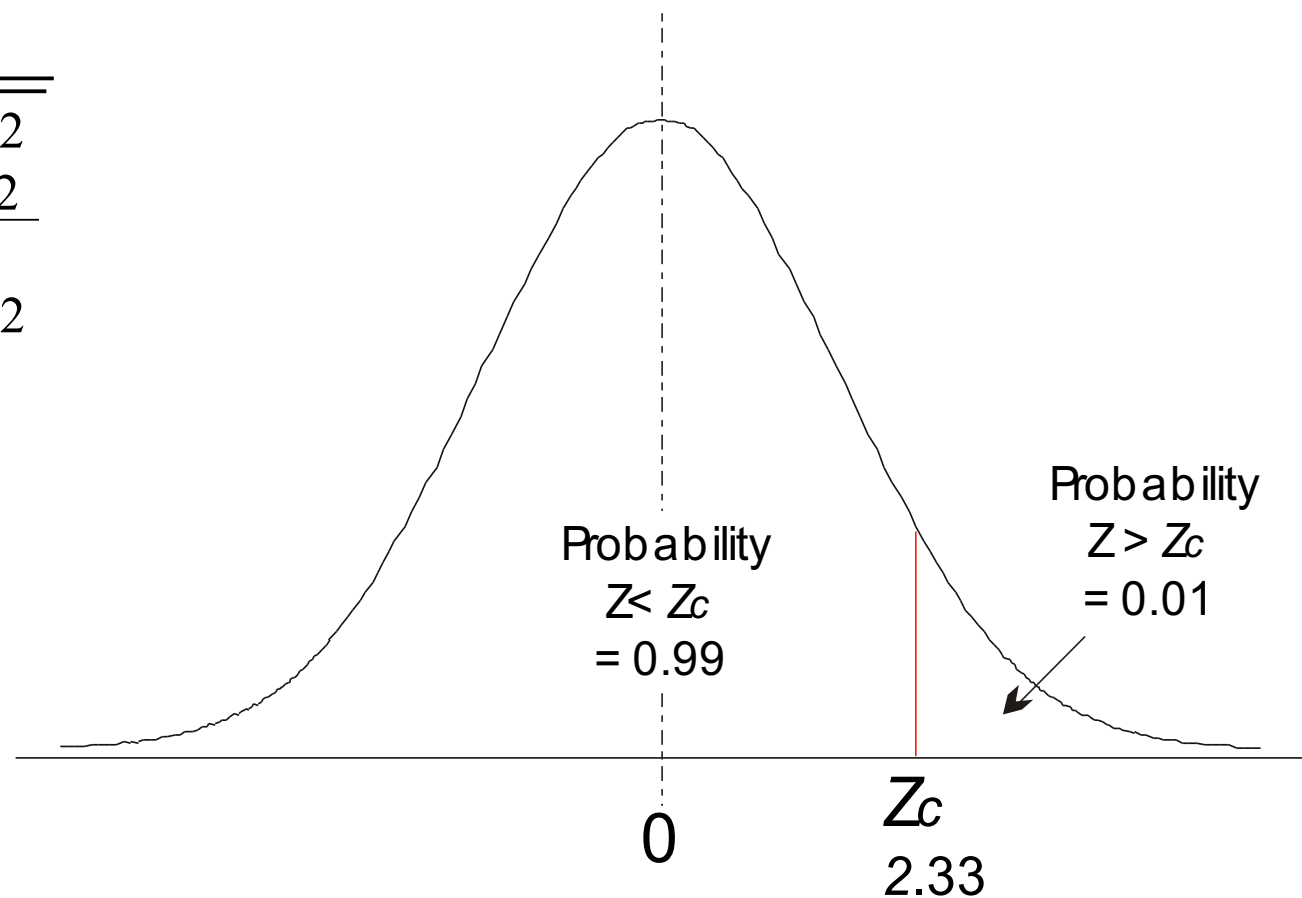
2. Decision: if test statistic to right of z_c then reject H_o

$\alpha=0.01: z_c = 2.33$



Showing Z: One-tailed

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$



Type I and Type II errors

- Since hypothesis testing is based on probability theory, the researcher cannot be completely certain and runs the risk of committing one of two types of errors:

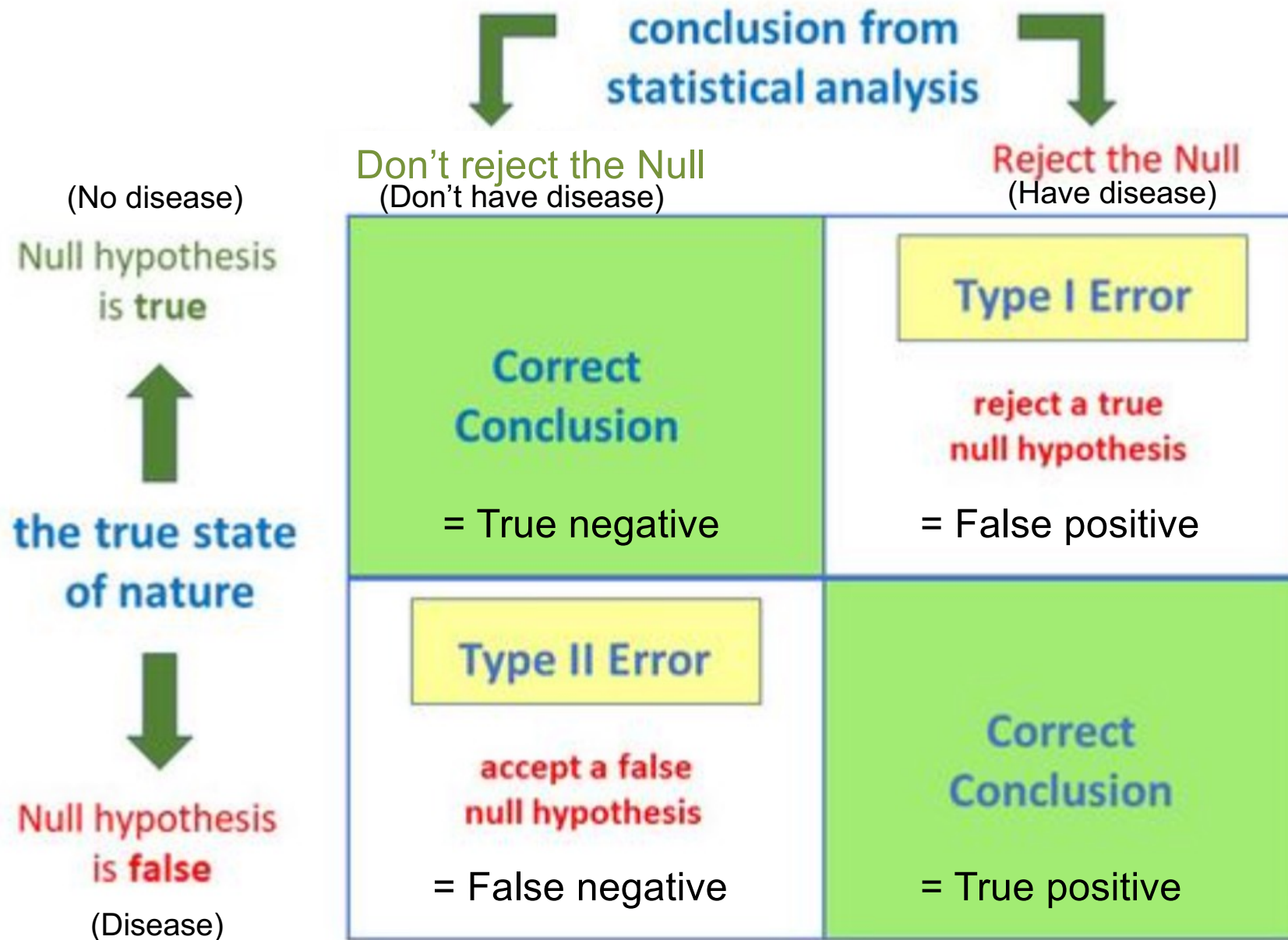
Type I error

- An error caused by rejecting the null hypothesis when it is true
- Probability of alpha (α)

Type II error

- An error caused by failing to reject the null hypothesis when the alternative hypothesis is true
- Probability of beta (β)

Type I and Type II errors



Lab 7

115.109 Topic 7 Lab Worksheet: Comparing Means of Composite Variables

Due date: October 5

Version: 1 (3.09)

Your Name:

Task 1: In the SustainabilityData sheet, construct the following composite

Columns to combine:

Societal Domain in cell CO5, enter: =average(BE5:BH5)

Environmental Domain in cell CP5, enter: =average(BI5:BL5)

Supplier Domain in cell CQ5, enter: =average(BM5:BO5)

Shareholder Domain in cell CR5, enter: =average(BP5:BR5)

Community Domain in cell CS5, enter: =average(BS5:BV5)

Employee Domain in cell CT5, enter: =average(BW5:BY5)

Customer Domain in cell CU5, enter: =average(BZ5:CC5)

Sustainability literature in cell CV5, enter: =average(CD5:CL5)

then copy down to row 1046

(nothing to show here, or verify: if you have done this task correctly, the yellow boxes in Task

CN	CO	CP	CQ	CR	CS	CT	CU	CV
	Societal Do	Environme	Supplier Do	Shareholde	Communit	Employee [	Customer [	Sustainabili
Which gender do you identify with?								
1: Male	1: Not at all	1: Not at all	1: Not at all	1: Not at all	1: Not at all	1: Not at all	1: Not at all	1: Strongly
2: Female	important	important	important	important	important	important	important	disagree
3: Gender diverse	5: Extremely Important	5: Extremely Important	5: Extremely Important	5: Extremely Important	5: Extremely Important	5: Extremely Important	5: Extremely Important	3: Neither disagree nor agree
								5: Strongly agree
Gender	Societal Domain	Environmental Domain	Supplier Domain	Shareholder Domain	Community Domain	Employee Domain	Customer Domain	Sustainability
2								

BE	BF	BG	BH
Societal Domain: How important is it that a ret			
Contributes to	Supports emp	Invests in you	Promotes div
1: Not at all important	1: Not at all important	1: Not at all important	1: Not at all important
5: Extremely Important	5: Extremely Important	5: Extremely Important	5: Extremely Important
Solve societal	Support employees	Youth	Diversity
3	3	3	3
3	4	4	3
4	4	5	5
3	3	3	3

BM	BN	BO
Supplier Domain		
Provide fair te	Communicate	Selects suppli
1: Not at all important	1: Not at all important	1: Not at all important
5: Extremely Important	5: Extremely Important	5: Extremely Important
Fair to suppliers	Communicate Suppliers	Ethical suppliers
4	4	4
4	3	4
5	5	5

Lab 7

Task 2: Generate means, standard deviations and sample sizes for the 'composite' variables constructed in Task 1, split by Gender

Process:

There are four different methods you **could** use to generate the required data (don't start these yet - read on!):

- Repeated PivotTables, using the Count, Average and StDev field value options;
- Repeated Filtering with the Subtotal commands for Count, Average and StDev.S (codes 2, 1 and 7 respectively)
- Calculating the descriptive statistics manually in sets, having sorted the data;
- Using Excel's Data Analysis Toolpak commands (which mostly require sorting).

☒ (Select All)

☒ 1

☐ 2

☐ 3

☐ 1: Male 2: Female 3: Gender diver

☒ Avg=1

☒ Count=2

☒ StDev=7

For this task we will use **method ii)**, calculated in cells CO1048-CV1050 of the SustainabilityData sheet, and Copy > Paste Values to the boxes beneath here.

- In SustainabilityData, activate the filter for Gender, selecting only 1 (Males), and also retain the Avg, StDev and Count rows
As per Lab 5 Task 3, calculate the **Subtotals** for CO5:CO1046 and the other columns up to CV5:CV1046 for the mean, count and std dev (using the =SUBTOTAL(1... etc function). {Use the area in CO10 Copy and Paste (as Values) the results in the blank space underneath (if you can see rows 1052-1054 use those, otherwise anywhere beneath the data, then remove the filter and paste again into row 1052-1054 for Males, and 1056-1058 for Females)
Repeat for Gender=2, and again copy the results beneath (rows 1056-1058 if possible). Transfer the results into the correct rows if needed (1052-1054 for Males, and 1056-1058 for Females)
- Copy and transpose the results below (Paste Special > Values and Transpose)
- Once complete, copy cells K33:M33 and Paste Special > Formulas into cells K34:K40 to copy down the Z-test calculations. Note any significant differences, and complete the interpretation box

Column	Composite Variable	Variable Mean	Males Variable Standard Deviation	Variable Sample Size	Females Variable Standard Deviation	Variable Sample Size	Numerator	Denominator	Z-test
CO	Societal Domain						0.00	#DIV/0!	#DIV/0!
CP	Environmental Domain								
CQ	Supplier Domain								
CR	Shareholder Domain								
CS	Community Domain								
CT	Employee Domain								
CU	Customer Domain								
CV	Sustainability literacy								

<<Z-test results Interpretation>>

$$Z = \frac{x_1 - x_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

What is the Z-test doing?

Null Hypothesis : Ho there is no difference in subsample means (eg male v female means)

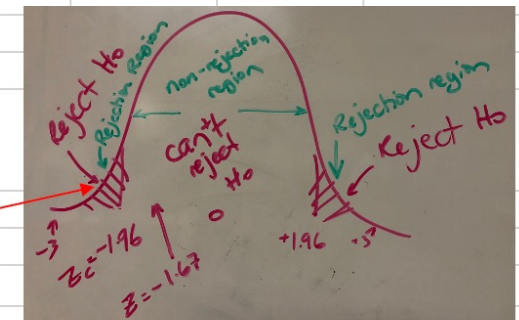
Alternative Hypothesis Ha : there is a difference in subsample means (2-tailed test is the default)
(one-tailed test: is one mean larger than the other)

Test statistic: see formula, compare to critical value of Z

Z_c ±1.96 (for α = 0.05, two-tailed)

Conclusion: are we in a rejection region?

NB: the Z test is for populations, or 'very large' samples.
Our sample is not very large, so we *should* be using a t-test



FIELD NAME

Search fields

Lab 7

42	Task 3: Repeat for Income, this time using PivotTables (method i) using the same groups constructed in Lab 6 (Lower, as groups 1+2); Middle (3+4) and Upper (5+6)									
43	1)	Create a PivotTable by Household Income2, with Values being Averages. Duplicate the finished table, and change the Values to StDev								
44		Unfortunately you have to change each variable from Count to Average (first table), and then from Average to StDev (second table) - there is no quick way to change them all (stupid Microsoft!)								
45		Copy result of PivotTables here, Averages first, using Paste Special > Values. Then COPY cells E50:E57 (averages for Middle) into G50, and F50:F57 into J50								
46	2)	Repeat for StDev, but in this case paste in rows 60 and below to avoid overwriting. COPY to cells E50:E57, H50:H57 and K50:K57.								
47	3)	Copy the sample size numbers into cells F50 and copy down, and repeat for I50 and L50. Then complete the sets of Z-statistic calculations. Highlight any significant differences and interpret								
48						Lower - Middle	Middle - Upper	Lower - Upper		
49	Column	Composite Variable	Variable Mean	Variable Standard Deviation	Variable Sample Size	Z-test	Z-test	Z-test	<<Z-test results interpretation>>	
50	CO	Societal Domain				#DIV/0!	#DIV/0!	#DIV/0!		
51	CP	Environmental Domain								
52	CQ	Supplier Domain								
53	CR	Shareholder Domain								
54	CS	Community Domain								
55	CT	Employee Domain								
56	CU	Customer Domain								
57	CV	Sustainability literacy								

$$Z = \frac{x_1 - x_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

What is the Z-test doing?

Null Hypothesis: H_0 there is no difference in subsample means (eg male v female means)

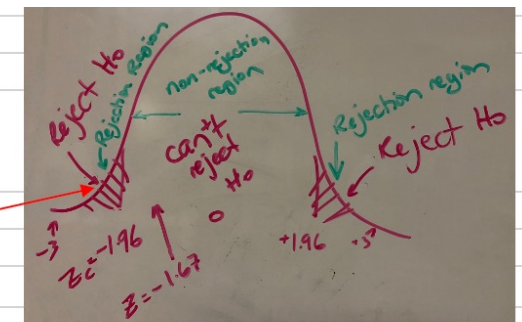
Alternative Hypothesis H_a : there is a difference in subsample means (2-tailed test is the default)
(one-tailed test: is one mean larger than the other)

Test statistic: see formula, compare to critical value of Z

Z_c ±1.96 (for $\alpha = 0.05$, two-tailed)

Conclusion: are we in a rejection region?

NB: the Z test is for populations, or 'very large' samples.
Our sample is not very large, so we *should* be using a t-test



FIELD NAME

Search fields

Lab 7

Task 4: Repeat Z-test, this time using the Excel Data Analysis Toolpak option, testing the composites by How Often Buy (column AY)

1) The Data Analysis Toolpak is turned off by default. For Mac: Tools menu > Excel Add-Ins > Analysis ToolPak
For Windows: File menu > Options > Add-ins > Excel Add-ins > Go > Analysis ToolPak
Once activated, you will see a Data Analysis group at the right of the Data tab ribbon. We will use the Z-test Two Sample for Means tool

2) The **WHOLE** dataset needs to be sorted by How Often Buy (column AY), smallest to largest, rows 5:1046 (ensure any Filters are turned off - Select All)
Write down the row numbers where HowOftenBuy subgroup 1+2+3 starts & ends, and where subgroup 4+5 starts & ends (since we don't include 6)
BUT: the Z-test tool requires knowing the Population variance for the two (sub)sample means we want to test.

3) So we need to know variance for each variable for Don't Buy Often (1,2&3) vs Often Buy (4+5). Repeat Task 2 above, this time Filtering for How Often Buy (column AY: options 1,2&3 vs 4&5)
Complete cells CO1060 to CV1066 of the SustainabilityData sheet and copy StDev here, and write down the variance values for Supplier Domain

4) Undo the Filter (ensure the data is sorted ascending by column AY: step 2 above). In the SustainabilityData sheet, in the Data ribbon, click the Data Analysis button and complete the Z-test dialog box like this (for columns CQ, then CT):

Column	Composite Variable	Standard deviation	Variance	Standard deviation	Variance	Copy results here:	z-Test: Two Sample for Means	Supplier Domain	z-Test: Two Sample for Means	Employee Domain
							Variable 1	Variable 2	Variable 1	Variable 2
CO	Societal Domain					Mean			Mean	
CP	Environmental Domain					Known Variance			Known Variance	
CQ	Supplier Domain					Observations			Observations	
CR	Shareholder Domain					Hypothesized Mean Difference			Hypothesized Mean Difference	
CS	Community Domain					z			z	
CT	Employee Domain					P(Z<=z) one-tail			P(Z<=z) one-tail	
CU	Customer Domain					z Critical one-tail			z Critical one-tail	
CV	Sustainability literacy					P(Z<=z) two-tail			P(Z<=z) two-tail	
						z Critical two-tail			z Critical two-tail	

Task 5: Create three line charts for the data (below the table), as follows:

Column	Composite Variable	Chart 1: Male vs Female		Chart 2: Lower vs Higher Income		Chart 3: Don't vs Often Buy	
		Male Mean	Female Mean	Lower Income Mean	Higher Income Mean	Don't Often Buy (1+2+3) Mean	Often Buy (4+5) Mean
CO	Societal Domain	0.00	0.00	0.00	0.00		
CP	Environmental Domain	0.00	0.00	0.00	0.00		
CQ	Supplier Domain	0.00	0.00	0.00	0.00		
CR	Shareholder Domain	0.00	0.00	0.00	0.00		
CS	Community Domain	0.00	0.00	0.00	0.00		
CT	Employee Domain	0.00	0.00	0.00	0.00		
CU	Customer Domain	0.00	0.00	0.00	0.00		
CV	Sustainability literacy	0.00	0.00	0.00	0.00		

Insert charts across from here: